1 Introduction

We now discuss integration of rational functions. The simplest nontrivial rational functions are of the form

$$R(x) = \frac{A}{ax+b}$$

for some constants A, a, b, with $A, a \neq 0$.

1. Evaluate $\int \frac{A}{ax+b}dx$.

2. Now consider the integral $\int \frac{2x-1}{x^2-x+6}dx$. Evaluate it using substitution.

The previous example involved a rather special rational function. A little thought shows that substitution is not going to work nicely in general. For example, if we try to evaluate $\int \frac{x+5}{x^2+x-2}dx$ by setting $u = x^2+x-2$, then $du = (2x+1)dx$, which does not match the numerator.

3. Verify that $\frac{2}{x-1} - \frac{1}{x+2} = \frac{x+5}{x^2+x-2}$ by finding a common denominator. Use your observation to evaluate $\int \frac{x+5}{x^2+x-2}dx$.

To evaluate $\int \frac{x+5}{x^2+x-2}dx$, we first expressed the integrand as a sum of simpler fractions, which were easy to integrate. This technique is called *partial fraction decomposition*. In the previous example we were simply given the partial fractions of the integrand. For this to be a useful technique of integration, we need to go in the opposite direction by answering the following questions:

(1) Given a rational expression, does it admit a partial fraction decomposition?

(2) If so, how do we find the partial fractions?

Let's begin by investigating a simple example.

4. Consider the rational expression $\frac{5x-13}{x^2-4x-5}$.

   (a) Factor the denominator.

   (b) Assume that there exist constants A, B such that

   $$\frac{5x-13}{x^2-4x-5} = \frac{A}{\text{factor 1}} + \frac{B}{\text{factor 2}},$$

   where 'factor 1' and 'factor 2' are the factors you found in part (a). Multiply both sides of the equation above by the product of these factors to get rid of the denominators.

   (c) Solve for A and B by plugging in each zero of x^2-4x-5 (one at a time) into your equation from part (b). Substitute the values of A and B into the original expression to obtain the partial fraction decomposition of $\frac{5x-13}{x^2-4x+5}$.

2 General Partial Fraction Decomposition

Consider now a general rational function

$$R(x) = \frac{P(x)}{Q(x)},$$

where $P(x)$ and $Q(x)$ are polynomials. If $\deg P < \deg Q$ then the rational function R is called *proper*. If the rational function is *improper*, that is, if $\deg P \geq \deg Q$, then we can divide Q into P (by polynomial long division) until a remainder $R(x)$ is obtained such that $\deg R < \deg Q$. Then

$$R(x) = \frac{P(x)}{Q(x)} = S(x) + \frac{R(x)}{Q(x)},$$

where $S(x)$ and $R(x)$ are polynomials, and $\frac{R(x)}{Q(x)}$ is proper.

As the next example shows, in some simple cases this preliminary step is all that is needed.

5. Evaluate $\displaystyle\int \frac{x^3+x}{x-1}dx$ by rewriting the integrand using long division.

Consider now a proper rational function $R(x) = \frac{P(x)}{Q(x)}$. The next step is to factor the denominator $Q(x)$ as far as possible. We will need the following theorem from algebra.

Theorem 2.1. *Any polynomial $Q(x)$ with real coefficients can be factored (over the reals) as a product of two types of factors:*

(i) linear factors (of the form $ax+b$)

(ii) irreducible quadratic factors (of the form ax^2+bx+c, where $b^2-4ac<0$).

Example 2.2. Consider $Q(x) = x^4 - 16$. Factoring $Q(x)$ as much as possible (over the reals) gives

$$Q(x) = (x^2-4)(x^2+4) = (x-2)(x+2)(x^2+4).$$

The factors $x-2$ and $x+2$ are linear, while x^2+4 is irreducible quadratic, since to factor it further would require introducing complex numbers as $x^2+4 = (x-2i)(x+2i)$.

With $Q(x)$ factored into linear and irreducible quadratic factors, the following theorem from algebra tells us that $R(x)$ admits a partial fraction decomposition and tells us how to find the partial fractions, answering Questions (1) and (2) in the previous section.

Theorem 2.3. *Let $R(x) = \frac{P(x)}{Q(x)}$ be a proper rational function, where the denominator $Q(x)$ has been factored into linear and irreducible quadratic factors. $R(x)$ can be written as a sum of partial fractions, where each factor in the denominator gives rise to terms in the partial fraction decomposition, as follows:*

- *A factor of the form $(ax+b)^k$ in the denominator gives rise to terms $\sum_{i=1}^{k} \frac{A_i}{(ax+b)^i}$ in the partial fraction decomposition.*
- *A factor of the form $(ax^2+bx+c)^k$ in the denominator gives rise to terms $\sum_{i=1}^{k} \frac{A_i x+B_i}{(ax^2+bx+c)^i}$ in the partial fraction decomposition.*

Example 2.4. Consider $R(x) = \dfrac{9x^7-12x^3+x-4}{(x-3)(2x+1)^3(2x^2+3x+5)^4}$.

- The factor $x-3$ gives rise to $\frac{A}{x-3}$.
- The factor $(2x+1)^3$ gives rise to $\frac{A_1}{2x+1} + \frac{A_2}{(2x+1)^2} + \frac{A_3}{(2x+1)^3}$.
- The factor $(2x^2+3x+5)^4$ gives rise to $\frac{C_1x+D_1}{2x^2+3x+5} + \frac{C_2x+D_2}{(2x^2+3x+5)^2} + \frac{C_3x+D_3}{(2x^2+3x+5)^3} + \frac{C_4x+D_4}{(2x^2+3x+5)^4}$.

The partial fraction decomposition is therefore has the form

$$\frac{9x^7-12x^3+x-4}{(x-3)(2x+1)^3(2x^2+3x+5)^4} = \frac{A}{x-3} + \frac{A_1}{2x+1} + \frac{A_2}{(2x+1)^2} + \frac{A_3}{(2x+1)^3} + \frac{C_1x+D_1}{2x^2+3x+5} + \frac{C_2x+D_2}{(2x^2+3x+5)^2} + \frac{C_3x+D_3}{(2x^2+3x+5)^3} + \frac{C_4x+D_4}{(2x^2+3x+5)^4}$$

6. Write out the general form of the partial fraction decomposition:

$$\frac{x^9-31x^6+17x-2}{(3x-2)^2(x^2+2x+1)(x^2+6x+7)^3} =$$

Once the general form of the partial fraction decomposition is given, it remains to solve for the coefficients appearing in numerators of the partial fractions. We have already seen an example of this in Question 4, where we were able to find these coefficients quickly by cleverly plugging in values for x. Here is another example which can be solved similarly.

7. Use partial fractions to evaluate the integral $\int \frac{x^2+4}{3x^3+4x^2-4x}dx$.

When the partial fraction decomposition has many terms, it is generally too hard to solve for the coefficients by just plugging in values of x. However, we can always reduce the problem to a system of linear equations, which can be solved systematically. The following example illustrates the general case.

Example 2.5. Consider $\int \frac{x^2-29x+5}{(x-4)^2(x^2+3)}dx$. The partial fraction decomposition of the integrand has the form

$$\frac{x^2-29x+5}{(x-4)^2(x^2+3)} = \frac{A}{x-4} + \frac{B}{(x-4)^2} + \frac{Cx+D}{x^2+3}.$$

Multiplying both sides by $(x-4)^2(x^2+3)$ gives

$$x^2-29x+5 = A(x-4)(x^2+3) + B(x^2+3) + (Cx+D)(x-4)^2.$$

While plugging in $x=4$ will give us B, it is not obvious how to find the other coefficients by plugging in values for x. Instead, multiply out the right hand side and collect like terms. This gives

$$x^2-29x+5 = (A+C)x^3 + (-4A+B-8C+D)x^2 + (3A+16C-8D)x - 12A+3B+16D.$$

For both sides to be equal, the coefficient of each power of x on both sides must be equal. [1] This implies that

$$\begin{aligned}
x^3: &\quad A+C = 0\\
x^2: &\quad -4A+B-8C+D = 1\\
x^1: &\quad 3A+16C-8D = -29\\
x^0: &\quad -12A+3B+16D = 5
\end{aligned}$$

This system can be solved by your favorite method (substitution, Gauss-Jordan Elimination, Cramer's Rule, multiply by an inverse matrix, ...) to find the unique solution

$$A = 1, B = -5, C = -1, D = 2.$$

Therefore

$$\begin{aligned}
\int \frac{x^2-29x+5}{(x-4)^2(x^2+3)}dx &= \int \left(\frac{1}{x-4} - \frac{5}{(x-4)^2} + \frac{-x+2}{x^2+3}\right)dx\\
&= \int \frac{dx}{x-4} - 5\int \frac{dx}{(x-4)^2} - \int \frac{x}{x^2+3}dx + 2\int \frac{dx}{x^2+3}\\
&= \boxed{\ln|x-4| + \frac{5}{x-4} - \frac{1}{2}\ln|x^2+3| + \frac{2}{\sqrt{3}}\tan^{-1}\left(\frac{x}{\sqrt{3}}\right) + C}.
\end{aligned}$$

[1] Proof: The equation $\sum_{i=1}^n c_i x^i = \sum_{i=1}^n d_i x^i$ is equivalent to $\sum_{i=1}^n (c_i - d_i)x^i = 0$. By the Fundamental Theorem of Algebra, a degree n polynomial has at most n roots. So for $\sum_{i=1}^n (c_i-d_i)x^i = 0$ to be true for all values of x, it must be that each coefficient is zero, that is, $c_i = d_i$ for all $i = 1, \ldots, n$. □

8. Evaluate $\int \frac{x^2+2x-1}{2x^3+3x^2-2x}dx$ using the method of the previous example.